

Counting Polygon Free Mosaics

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Abstract

Imagine you are tasked with tiling a rectangular bathroom floor that is m units by n units, where m and n are positive integers. At your disposal is an unending supply of 7 distinct types of tiles. These tile types are composed of unit squares with line segments connecting a pair of sides at their midpoints, as well as the “blank” tile with no line segment. Some of these floors, referred to as mosaics in the literature, can contain polygons formed by the line segments. Hong and Oh gave bounds on the number of polygon mosaics, which are mosaics in which any tile that is not part of a polygon is the blank tile. We exactly count polygon mosaics. We also count the more difficult polygon free mosaics, which are mosaics that do not contain a polygon.

1 Mosaics

Imagine you are tasked with tiling a rectangular bathroom floor that is m units by n units, where m and n are positive integers. At your disposal is an unending supply of 7 distinct types of tiles. These tile types, diagrammed in Figure 1, are composed of unit squares with lines segments connecting 2 sides at their midpoints, as well as the “blank tile”.

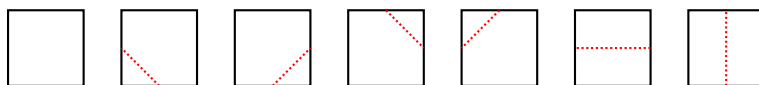
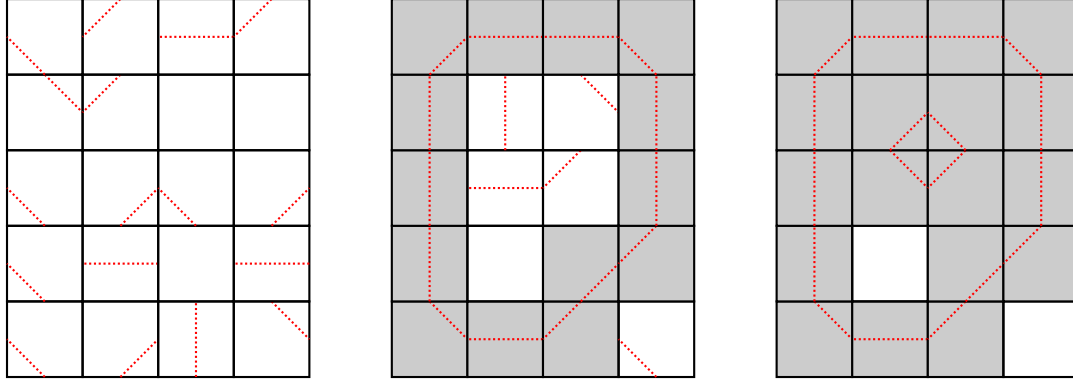


Figure 1: The tile set

The task is complete once you place mn tiles to cover the floor. We call a fully tiled $m \times n$ floor an (m, n) *mosaic*. Figure 2 shows various $(5, 4)$ mosaics.

We are interested in the number of mosaics in which the line segments do not form any polygons, as well as the number of mosaics in which any tile that is not part of a polygon is the blank tile. We highlight in gray the tiles that are part of a polygon in Figure 2.

The types of polygons that can be formed from the tiles are more commonly known as “self-avoiding polygons” in the literature [2, 3] to highlight their connection with self-avoiding walks. As self-avoiding polygons are formed by walks on the square lattice, some authors [4] have chosen to represent the tiles connecting adjacent sides with two line segments that meet at the center of the square at a right angle, as in Figure 3. We choose our construction for visual clarity, but point out the results of this paper are true for either style of tile.



(a) A polygon free mosaic (b) A mosaic that is not polygon free (c) A polygon mosaic

Figure 2: Examples of $(5, 4)$ mosaics

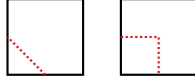


Figure 3: Our tile style (left) and another common style [4] (right)

Definition 1.1. An (m, n) *polygon free mosaic* is an (m, n) mosaic that does not contain a polygon. An (m, n) *polygon mosaic* is an (m, n) mosaic such that any tile that is not part of a polygon is the blank tile.

Figure 2a depicts a polygon free mosaic, Figure 2b depicts a mosaic that is neither a polygon free mosaics nor a polygon mosaic, and Figure 2c depicts a polygon mosaic. Notice that for a given m and n , the only mosaic that is both a polygon mosaic and a polygon free mosaic is the mosaic of all blank tiles. The goals of this paper are twofold, counting polygon mosaics and counting polygon free mosaics.

2 Mapping Mosaics to Proper Lattices

For positive integers m and n , an (m, n) *binary lattice* is a rectangular lattice of $m + 1$ by $n + 1$ vertices, with each vertex labeled 0 or 1. We also define a *framed lattice* to be a binary lattice in which all boundary vertices are labeled 0. An example of a $(5, 4)$ framed lattice is shown on the right of Figure 4. Let a *cell* be a $(1, 1)$ binary lattice. The sixteen possible cells are:

$$\begin{matrix} 0 & 0 \\ 0 & 0 \end{matrix}, \begin{matrix} 0 & 0 \\ 0 & 1 \end{matrix}, \begin{matrix} 0 & 0 \\ 1 & 0 \end{matrix}, \begin{matrix} 0 & 0 \\ 1 & 1 \end{matrix}, \begin{matrix} 0 & 1 \\ 0 & 0 \end{matrix}, \begin{matrix} 0 & 1 \\ 0 & 1 \end{matrix}, \begin{matrix} 0 & 1 \\ 1 & 0 \end{matrix}, \begin{matrix} 0 & 1 \\ 1 & 1 \end{matrix}, \begin{matrix} 1 & 0 \\ 0 & 0 \end{matrix}, \begin{matrix} 1 & 0 \\ 0 & 1 \end{matrix}, \begin{matrix} 1 & 0 \\ 1 & 0 \end{matrix}, \begin{matrix} 1 & 0 \\ 1 & 1 \end{matrix}, \begin{matrix} 1 & 1 \\ 0 & 0 \end{matrix}, \begin{matrix} 1 & 1 \\ 0 & 1 \end{matrix}, \begin{matrix} 1 & 1 \\ 1 & 0 \end{matrix}, \text{ and } \begin{matrix} 1 & 1 \\ 1 & 1 \end{matrix}.$$

A *proper lattice* is a framed lattice that does not contain any $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ or $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$ cells.

Let P be a polygon mosaic. We will create a binary lattice from P by replacing every point that is at a corner of a tile of P by either 0 or 1. For each such point, we count

the number of polygons of P that surround it, i.e., polygons for which the point is in the polygon's interior. If this number is even, label the vertex 0, and if it is odd, label the vertex 1. We then delete all dotted lines that are part of polygons of P . We define $\mathcal{L}(P)$ to be the resulting binary lattice. See Figure 4 for an example. If P is an (m, n) polygon mosaic, i.e., m tiles high and n tiles wide, then $\mathcal{L}(P)$ is an (m, n) binary lattice, i.e., $m + 1$ vertices high and $n + 1$ vertices wide. In fact, $\mathcal{L}(P)$ is an (m, n) framed lattice because every point on the edge of P is surrounded by no polygons.

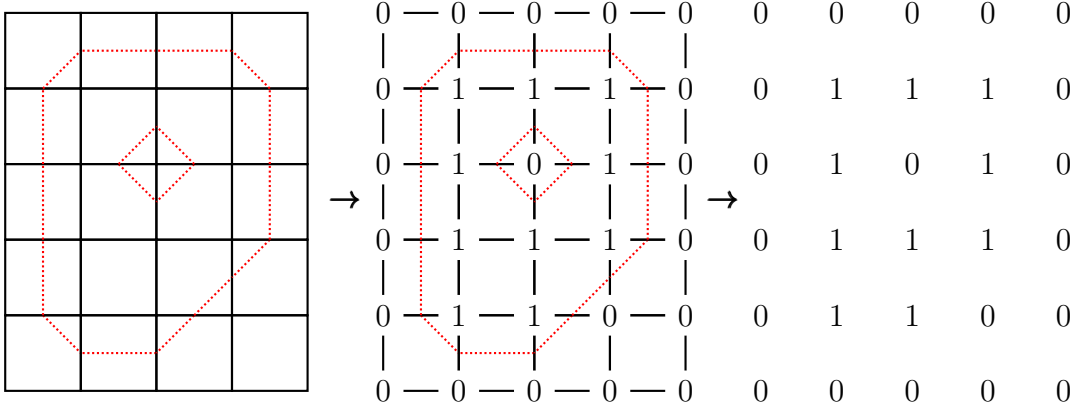


Figure 4: A polygon mosaic P (right), its image $\mathcal{L}(P)$ (right), and the two superimposed (middle).

In our definition of $\mathcal{L}$, a tile of a polygon mosaic P corresponds to a cell of $\mathcal{L}(P)$. For a tile of P that is not the blank tile, consider laying its corresponding cell over it, as in the middle of Figure 4. The dotted line of the tile separates the vertices of the cells, with vertices labeled 0 on one side and labeled 1 on the other. For example, the second from left tile of Figure 1 must map to either the cell $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$ or the cell $\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}$. Which one? Consider first where the interior of this polygon is relative to this dotted line, and second, the number of other polygons that contain this polygon. For example, the tile maps to $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$ if the interior is below the dotted line and an even number of polygons contain this polygon. On the other hand, a blank tile of P corresponds to a cell that is either $\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}$ or $\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}$. With seven tiles, and no cell able to be the image of distinct tiles, $\mathcal{L}(P)$ can contain fourteen types of cells. The two types that never appear in $\mathcal{L}(P)$ are $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$. That is, $\mathcal{L}(P)$ is proper.

Given any proper lattice ℓ , we can reverse this process, creating a mosaic by replacing each cell by the unique tile that can map to it. Define $\mathcal{L}^{-1}$ by restricting this map to proper lattices, and let us show that $\mathcal{L}^{-1}$ is indeed an inverse to $\mathcal{L}$, thus proving that $\mathcal{L}$ gives a bijection from (m, n) polygon mosaics to (m, n) proper lattices. Our discussion above shows that $\mathcal{L}^{-1} \circ \mathcal{L}(P) = P$ for any polygon mosaic P , so fixing a proper lattice ℓ , we must show that $\mathcal{L}^{-1}(\ell)$ is a polygon mosaic and that $\mathcal{L} \circ \mathcal{L}^{-1}(\ell) = \ell$.

First observe that polygon mosaics can be characterized as those mosaics in which every end of a dotted line on a tile meets, on an adjacent tile, another end of a dotted line. In the mosaic $\mathcal{L}^{-1}(\ell)$, these ends are exactly the points which, in the corresponding cell of ℓ , lie halfway between adjacent vertices of ℓ , one labeled 0 and the other labeled 1. Because ℓ is a framed lattice, the vertex labeled 1 is an internal vertex. Thus, another cell of ℓ shares

these two vertices, and the dotted lines of $\mathcal{L}^{-1}(\ell)$ from the two corresponding tiles meet. Therefore $\mathcal{L}^{-1}(\ell)$ is a polygon mosaic.

By our analysis of $\mathcal{L}$ and our definition of $\mathcal{L}^{-1}$, each cell of $\mathcal{L} \circ \mathcal{L}^{-1}(\ell)$ must either be equal to the corresponding cell of ℓ or is formed from the corresponding cell by relabeling all vertices 0 as 1 and 1 as 0. Furthermore, since the cells are connected, i.e., none is isolated, we must then have that $\mathcal{L} \circ \mathcal{L}^{-1}(\ell)$ itself is either equal to ℓ or formed by relabeling all vertices 0 as 1 and 1 as 0. By hypothesis, ℓ is a proper lattice, and thus a framed lattice, and by construction, every binary lattice in the range of $\mathcal{L}$ is a framed lattice. Therefore $\mathcal{L} \circ \mathcal{L}^{-1}(\ell) = \ell$. This concludes our proof that $\mathcal{L}^{-1}$ is the inverse of $\mathcal{L}$ and thus that $\mathcal{L}$ is bijective.

3 Counting Polygon Mosaics

Definition. A *cell product function* is a real valued function whose domain is the set of all binary lattices and whose value on any lattice is the product of its values on every cell in the lattice.

We will show that certain cell product functions allow us to count both polygon mosaics (Section 3) and polygon free mosaics (Section 4), as well as certain variations we will create by varying the tile set (Section 6).

Theorem 3.1. *Let v be the cell product function that maps the cells $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$ to 0, and maps the fourteen other cells to 1. Then for all positive integers m and n , the number of (m, n) polygon mosaics equals $\sum_{\ell} v(\ell)$, where the sum is over all (m, n) framed lattices ℓ .*

Proof. For any framed lattice ℓ , $v(\ell)$ is 1 if ℓ is proper, and otherwise is 0, so that $\sum_{\ell} v(\ell)$ counts the number of proper (m, n) framed lattices. Our conclusion now follows from the fact that $\mathcal{L}$ gives a bijection from the set of (m, n) polygon mosaics to the set of proper (m, n) framed lattices. $\square$

For any cell product function v , we can use matrices to more efficiently compute the sum $\sum v(\ell)$, where the sum is over all (m, n) framed lattices ℓ .

Theorem 3.2. *Let v be any cell product function. Define the 1×1 matrices $A_1 = (v(\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}))$, $B_1 = (v(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}))$, $C_1 = (v(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}))$, and $D_1 = (v(\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}))$, and for positive integers n , define the $2^n \times 2^n$ matrices*

$$\begin{aligned} A_{n+1} &= \begin{pmatrix} v(\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix})A_n & v(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix})B_n \\ v(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix})C_n & v(\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix})D_n \end{pmatrix} & B_{n+1} &= \begin{pmatrix} v(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix})A_n & v(\begin{smallmatrix} 0 & 0 \\ 1 & 1 \end{smallmatrix})B_n \\ v(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix})C_n & v(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix})D_n \end{pmatrix} \\ C_{n+1} &= \begin{pmatrix} v(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix})A_n & v(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix})B_n \\ v(\begin{smallmatrix} 1 & 1 \\ 0 & 0 \end{smallmatrix})C_n & v(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix})D_n \end{pmatrix} & D_{n+1} &= \begin{pmatrix} v(\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix})A_n & v(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix})B_n \\ v(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix})C_n & v(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix})D_n \end{pmatrix}. \end{aligned}$$

Then for all positive integers m and n , the upper left entry of A_n^m equals $\sum_{\ell} v(\ell)$, where the sum is over all (m, n) framed lattices ℓ .

Before we prove Theorem 3.2, we set up some notation and prove Proposition 3.3. Throughout this section, v and the matrices A_n, B_n, C_n , and D_n are as in Theorem 3.2. Let $\beta_n(k)$ be the n digit binary representation of the nonnegative integer $k < 2^n$, with $\beta_0(0)$ being the empty string. For example, since v is a cell product function,

$$v \begin{pmatrix} 0 & \beta_2(1) & 0 \\ 0 & \beta_2(0) & 0 \end{pmatrix} = v \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = v \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} v \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

For $m \times n$ matrices we number the rows $0, 1, \dots, m-1$ top to bottom, and columns $0, 1, \dots, n-1$ left to right.

Proposition 3.3. *For a positive integer n and nonnegative integers $i, j < 2^{n-1}$, the (i, j) -th entry of*

$$\begin{aligned} A_n \text{ is } v \begin{pmatrix} 0 & \beta_{n-1}(i) & 0 \\ 0 & \beta_{n-1}(j) & 0 \end{pmatrix}, \quad B_n \text{ is } v \begin{pmatrix} 0 & \beta_{n-1}(i) & 0 \\ 1 & \beta_{n-1}(j) & 0 \end{pmatrix}, \\ C_n \text{ is } v \begin{pmatrix} 1 & \beta_{n-1}(i) & 0 \\ 0 & \beta_{n-1}(j) & 0 \end{pmatrix}, \quad D_n \text{ is } v \begin{pmatrix} 1 & \beta_{n-1}(i) & 0 \\ 1 & \beta_{n-1}(j) & 0 \end{pmatrix}. \end{aligned}$$

Proof. We induct on n . The $n = 1$ case is trivial, as $\beta_0(0)$ is the empty string. We next assume our result for some fixed $n \geq 1$. The induction step requires proving the proposition for entries in $A_{n+1}, B_{n+1}, C_{n+1}$, and D_{n+1} . As these matrices are defined by quadrants of the matrices A_n, B_n, C_n , and D_n , there are sixteen cases to examine. We do two of the sixteen cases for illustration, that being the upper right quadrant of A_{n+1} and then the lower left quadrant of D_{n+1} .

Let i, j be nonnegative integers less than 2^{n-1} . For the upper right block matrix in A_{n+1} , by definition the $(i, 2^{n-1} + j)$ entry of A_{n+1} equals $v \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ times the (i, j) entry of B_n . By our induction hypothesis, this equals

$$v \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} v \begin{pmatrix} 0 & \beta_{n-1}(i) & 0 \\ 1 & \beta_{n-1}(j) & 0 \end{pmatrix} = v \begin{pmatrix} 0 & 0 & \beta_{n-1}(i) & 0 \\ 0 & 1 & \beta_{n-1}(j) & 0 \end{pmatrix} = v \begin{pmatrix} 0 & \beta_n(i) & 0 \\ 0 & \beta_n(2^{n-1}+j) & 0 \end{pmatrix},$$

as desired. Similarly, for the lower left quadrant of D_{n+1} , by definition the $(2^{n-1} + i, j)$ entry of D_{n+1} equals $v \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ times the (i, j) entry of C_n . By our induction hypothesis, this equals

$$v \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} v \begin{pmatrix} 1 & \beta_{n-1}(i) & 0 \\ 0 & \beta_{n-1}(j) & 0 \end{pmatrix} = v \begin{pmatrix} 1 & 1 & \beta_{n-1}(i) & 0 \\ 1 & 0 & \beta_{n-1}(j) & 0 \end{pmatrix} = v \begin{pmatrix} 1 & \beta_n(2^{n-1}+i) & 0 \\ 1 & \beta_n(j) & 0 \end{pmatrix},$$

again as desired. These arguments must be adapted to prove the desired results for all quadrants of $A_{n+1}, B_{n+1}, C_{n+1}$, and D_{n+1} , but all the work is similar. $\square$

Proof of Theorem 3.2. Fix n . We first introduce some terminology. For all positive integers m , we say a *height m strip* is an (m, n) binary lattice in which all vertices in the farthest left and farthest right columns are labeled zero. Then for a height m strip ℓ and a nonnegative integer $i < 2^{n-1}$, we say a particular row of ℓ is *determined by i* if that row is equal to $0 \beta_{n-1}(i) 0$. Note that an (m, n) framed lattice is exactly a height m strip in which both the top and bottom rows are determined by 0. Thus, to prove our theorem it suffices to show the following.

Claim. For all positive integers m and nonnegative integers $i, j < 2^{n-1}$, the (i, j) entry of A_n^m equals $\sum v(\ell)$, where the sum is over all height m strips ℓ whose first row is determined by i and whose last row is determined by j .

To prove the claim, first observe that for each i and j , there exists a unique height 1 strip whose first row is determined by i and whose last row is determined by j . By Proposition 3.3, v applied to this strip is equal to the (i, j) entry of A_n . This proves our claim when $m = 1$. We continue by induction, so for some $m \geq 2$, assume the claim for $m - 1$.

Fix nonnegative integers $i, j < 2^{n-1}$, and consider some nonnegative integer $k < 2^{n-1}$. As in the $m = 1$ case, Proposition 3.3 tells us that the (i, k) entry of A_n equals v applied to the unique height 1 strip with first and last rows determined respectively by i and by k . By our induction hypothesis, the (k, j) entry of A_n^{m-1} equals $\sum v(\ell)$, where the sum is over all height $m - 1$ strips ℓ whose first and last rows are determined respectively by k and by j . Consider applying v both to the height 1 strip indicated above and to one of the height $m - 1$ strips ℓ in this sum. Because v is a cell product function, the product of these two numbers is equal to v applied to the height m strip formed by overlapping the bottom row of the height 1 strip to the top row of ℓ , which we illustrate as

$$v \begin{pmatrix} 0 & \beta_{n-1}(i) & 0 \\ 0 & \beta_{n-1}(k) & 0 \end{pmatrix} v \begin{pmatrix} 0 & \beta_{n-1}(k) & 0 \\ \vdots & \vdots & \vdots \\ 0 & \beta_{n-1}(j) & 0 \end{pmatrix} = v \begin{pmatrix} 0 & \beta_{n-1}(i) & 0 \\ 0 & \beta_{n-1}(k) & 0 \\ \vdots & \vdots & \vdots \\ 0 & \beta_{n-1}(j) & 0 \end{pmatrix}.$$

From this we see that the product of the (i, k) entry of A_n and the (k, j) entry of A_n^{m-1} is equal to $\sum v(\ell)$, where the sum is now over all height m strips ℓ whose first, second, and last rows are determined respectively by i , by k , and by j . This tells us, in turn, that when we additionally sum over all nonnegative $k < 2^{n-1}$, we can conclude that the (i, j) entry of $A_n \cdot A_n^{m-1} = A_n^m$ is, as desired, the sum $\sum v(\ell)$ where the sum is over all height m strips ℓ whose first and last rows are determined respectively by i and by j . $\square$

Substituting the values for v from Theorem 3.1 into the matrices from Theorem 3.2 gives the following theorem.

Theorem 3.4. *Define the 1×1 matrices $A_1 = B_1 = C_1 = D_1 = (1)$, and for positive integers n , define the $2^n \times 2^n$ matrices*

$$\begin{aligned} A_{n+1} &= \begin{pmatrix} A_n & B_n \\ C_n & D_n \end{pmatrix} & B_{n+1} &= \begin{pmatrix} A_n & B_n \\ 0C_n & D_n \end{pmatrix} \\ C_{n+1} &= \begin{pmatrix} A_n & 0B_n \\ C_n & D_n \end{pmatrix} & D_{n+1} &= \begin{pmatrix} A_n & B_n \\ C_n & D_n \end{pmatrix}. \end{aligned}$$

Then the number of (m, n) polygon mosaics is the upper left entry of A_n^m .

Now that we have counted polygon mosaics, we move to the more complicated polygon free mosaics.

4 Counting Polygon Free Mosaics

The main goal of this section is to prove the following theorem.

Theorem 4.1. *Let v be the cell product function that maps the cells $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$ to 0, $\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}$ to 7, $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}$ to -1 , and the ten other cells to 1. Then for all positive integers m and n , the number of (m, n) polygon free mosaics equals $\sum_{\ell} v(\ell)$, where the sum is over all (m, n) framed lattices ℓ .*

Throughout this section, let v be as in Theorem 4.1. For example, if ℓ is the proper lattice on the right of Figure 4, we have $v(\ell) = 7^2$, as there is one $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$ cell, one $\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}$ cell, one $\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}$ cell, and one $\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}$ cell. For the proof of Theorem 4.1, we need to following critical result.

Proposition 4.2. *For a polygon mosaic P , let $\ell = \mathcal{L}(P)$. If P contains an even [resp. odd] number of polygons then $v(\ell)$ is positive [resp. negative].*

We might want to think of this result as relating a “global” characteristic and “local” characteristic, since the number of polygons in P is determined by looking at the entire mosaic, whereas $v(\ell)$ is calculated by looking at one cell at a time. The reader may wish to compare this to $\mathcal{L}$, which was defined “globally” versus $\mathcal{L}^{-1}$, which was defined “locally.”

Proof of Proposition 4.2. Let N be the number of vertices of ℓ that are 1, and define ℓ_0 to be the lattice formed by changing these N vertices from 1 to 0. That is, ℓ_0 has the same dimensions as ℓ but all vertices are 0. We will change ℓ_0 back to ℓ by flipping these vertices back from 0 to 1, one vertex at a time. The order in which we flip vertices is as follows.

We flip vertices one row at a time, working from top to bottom. In each row we make two passes, each time moving from left to right, and each time flipping a vertex if the corresponding vertex of ℓ is 1. On the first pass we flip a vertex if the vertex directly above it is 1. (Because we are working the rows from top to bottom, this vertex is 1 in both our current lattice and also in ℓ .) On the second pass we flip the remaining vertices of the row, i.e., if the vertex directly above it is 0.

This procedure creates a sequence $\ell_0, \ell_1, \dots, \ell_N$ of lattices with $\ell_N = \ell$. Because $\ell = \mathcal{L}(P)$, we know ℓ is a proper lattice. Since every ℓ_i has the same dimensions as ℓ and has vertices of 0 in at least the same positions as in ℓ , we see every ℓ_i is a framed lattice. We will prove that every ℓ_i is also a proper lattice (which is the reason we make two passes through each row) and that by taking $P_i = \mathcal{L}^{-1}(\ell_i)$, the proposition holds for P_i and ℓ_i , for all i . Since $\ell = \ell_N$, this will complete our proof.

In the transition from ℓ_{i-1} to ℓ_i , because every ℓ_i is framed, the single vertex that changes is adjacent to eight vertices, as in Figure 5, where the changing vertex $\#$ is 0 in ℓ_{i-1} and 1 in ℓ_i , and the remaining vertices are labeled for their relative position to this central vertex, such as dl for down left and u for up.

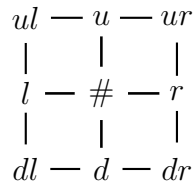


Figure 5: Cases Diagram

We need to consider all possible adjacent vertices. Since our process goes from top to bottom one row at a time, we have $(dl, d, dr) = (0, 0, 0)$. Also, because of the way in which we make our two passes, if $u = 1$ then $r = 0$ and $(ul, l) \neq (0, 1)$, whereas if $u = 0$ then $(ur, r) \neq (0, 1)$. Additionally, suppose $u = 0$. Then neither (ul, l) nor (ur, r) equals $(1, 0)$. Indeed, if $(ul, l) = (1, 0)$ then since we are making the second pass through the row, we would have a cell $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$ in ℓ_i and thus also in $\ell_N = \ell$, contradicting that ℓ is a proper lattice. Similarly, $(ur, r) = (1, 0)$ implies that ℓ has a cell $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ (since with $ur = 1$, r cannot change in the second pass), again contradicting that ℓ is a proper lattice. Putting all this together, on the first pass, i.e., when $u = 1$, the vertices can only be as in the six cases pictured in Figure 6. Similarly, on the second pass we have $u = 0$ and one of the six cases of Figure 7.

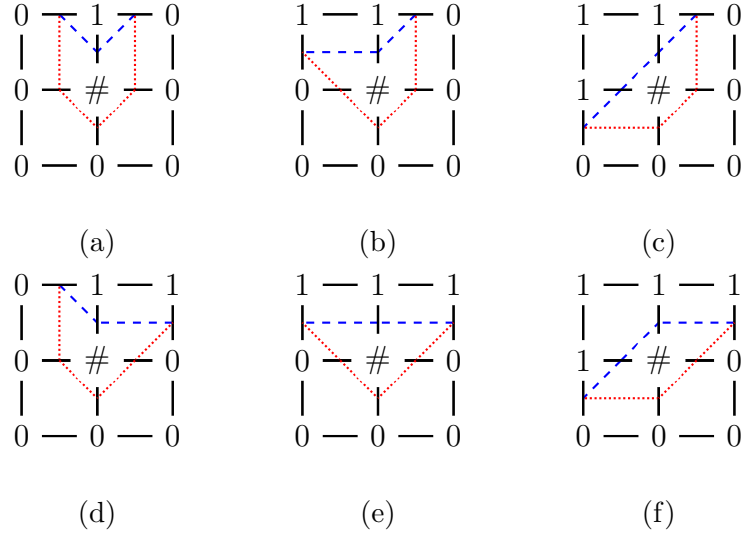


Figure 6: When $u = 1$

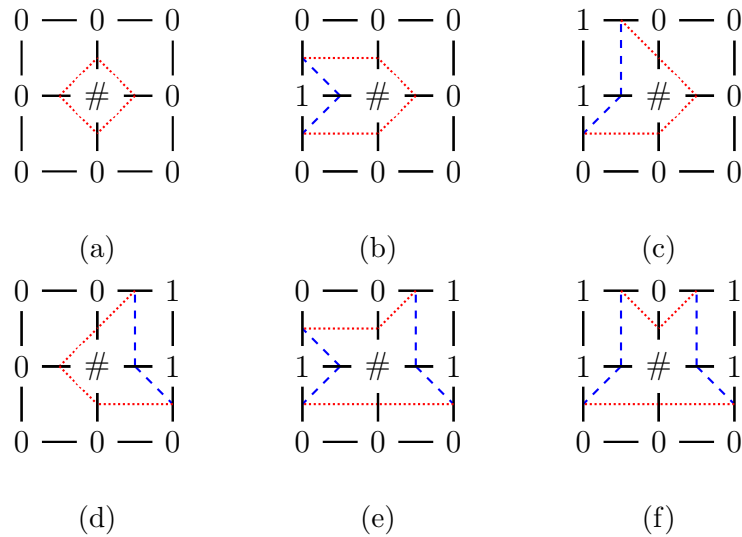


Figure 7: When $u = 0$

Since trivially ℓ_0 is a proper lattice, the twelve cases in Figures 6 and 7 show inductively that every ℓ_i is a proper lattice because none of the cases create a $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ or $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$ cell when $\#$ changes from 0 to 1. For every i , we have already seen that ℓ_i is a framed lattice, so let $P_i = \mathcal{L}^{-1}(\ell_i)$. Figures 6 and 7 also show the portions of P_{i-1} and P_i that are in these two cells, where the dashed lines are from the polygons of P_{i-1} and the dotted lines from the polygons of P_i . (Recall that $\#$ is 0 in ℓ_{i-1} and 1 in ℓ_i .)

Let us say a cell is *negative* if it is $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$ or $\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}$, i.e., if its evaluation under v is negative. The conclusion of Proposition 4.2 can be rephrased by saying that the number of negative cells in ℓ and the number of polygons in P have the same parity, i.e., are both even or both odd. For example, ℓ_0 satisfies the proposition because ℓ_0 has no negative cells and P_0 has no polygons.

Assume now that some P_{i-1} and ℓ_{i-1} satisfies the proposition, and let us show that the same is true for P_i and ℓ_i . This will complete the proposition. We know that we are in one of the twelve cases of Figures 6 and 7, and we must look at the number of negative cells in ℓ_i versus ℓ_{i-1} , and the number of polygons in P_i versus P_{i-1} . An easy case is (a) of Figure 7 because ℓ_i has one more negative cell (specifically, a cell $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$), and P_i has one more polygon than P_{i-1} .

Let us now look in the remaining cases at how the number of negative cells changes from ℓ_{i-1} to ℓ_i . In the six cases of Figure 6 and also case (d) of Figure 7, the four pictured cells have no negative cells regardless of whether $\#$ equals 0 or 1, so there is no change. The number is also unchanged in (b) of Figure 7 because a single negative cell $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$ changes position between ℓ_{i-1} and ℓ_i . In (c) of Figure 7, the number of negative cells increases by 2 because changing 0 to 1 in the central position creates both a cell $\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}$ and a cell $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$. Finally, the number decreases by 1 in (e) of Figure 7 and increases by 1 in (f) of Figure 7. Therefore, we can complete the theorem by showing that in the six cases of Figure 6 and in (b), (c), (d) of Figure 7, P_{i-1} and P_i have the same number of polygons, whereas in (e) and (f) of Figure 7, the number of polygons in P_{i-1} and P_i differs by 1.

Consider, for example, (c) of Figure 6, and look just above the lower left corner and just left of the upper right corner. In P_{i-1} these are connected by a straight dashed line, and in P_i these are connected by dotted line segments. We call both of these *inside connections*. Because P_{i-1} and P_i are polygon mosaics, these same points are connected by straight line segments outside the pictured area, and we will call these *outside connections* because they connect points on the edge of the pictured area with a path outside the area. This outside connection is the same in P_{i-1} and P_i . Therefore P_{i-1} and P_i have the same number of polygons because the polygon with this outside connection changes shape, and all other polygons are unchanged. Similar reasoning completes the remaining cases of Figure 6 and cases (b), (c), (d) of Figure 7.

We now must show that in (e) and (f) of Figure 7, the number of polygons in P_{i-1} and P_i differ by 1. We will explicitly show this in (e), and essentially identical reasoning applies to (f). There are four places where inside connections meet the edge of the pictured area. Beginning in the lower left and working clockwise, let x_1, x_2, x_3 , and x_4 be these points. For example, x_3 is just left of the upper right corner and x_4 is just above the lower right corner. In P_{i-1} , there is an inside connection between x_1 and x_2 , and another between x_3 and x_4 , whereas P_i has inside connections between x_1 and x_4 , and between x_2 and x_3 .

Any outside connections in P_{i-1} and P_i must be identical, and since these are polygon

mosaics, each of x_1, x_2, x_3 , and x_4 is the end of some outside connection. One possibility is that we have one outside connection between x_1 and x_2 , and another between x_3 and x_4 . In this case, the inside and outside connections of P_{i-1} complete two polygons, one including x_1 and x_2 , and the other including x_3 and x_4 . However, P_i will have one fewer polygon because a single polygon contains all x_i . Another possibility is to have one outside connection between x_1 and x_4 , and another between x_2 and x_3 . In this case, in P_{i-1} a single polygon includes all x_i , whereas in P_i they complete two polygons, one containing x_1 and x_4 , and the other containing x_2 and x_3 , and therefore P_i has one more polygon than P_{i-1} . These are the only possibilities since it is impossible to have an outside connection between x_1 and x_3 , and another between x_2 and x_4 . This is because our polygons are in a plane, and polygons in a mosaic are not able to cross one another. We have completed our proof. $\square$

Proof of Theorem 4.1. Fixing m and n , throughout this proof we use “polygon mosaic” for “ (m, n) polygon mosaic,” “framed lattice” for “ (m, n) framed lattice,” and so on.

We introduce the following identification for polygon mosaics. Let $\{P_i\}_{i \in J}$ be the set of all polygon mosaics that have exactly one polygon, and for any subset I of J , let P_I , if it exists, be the polygon mosaic that contains exactly the polygons of P_i for all $i \in I$. For example, $P_\emptyset$ is the polygon mosaic containing no polygons, i.e., the mosaic entirely composed of blank tiles, and for any $i \in J$, $P_{\{i\}}$ always exists and equals P_i . For an arbitrary $I \subseteq J$, P_I exists if and only if the polygons of P_i , for $i \in I$, do not “overlap.” More precisely, P_I exists if and only if for every mosaic location, there is at most one $i \in I$ such that P_i has a non-blank tile there. Notice that every polygon mosaic can be uniquely written as P_I for some $I \subseteq J$. For example, when $(m, n) = (4, 2)$, we can take $J = \{1, 2, 3, 4, 5, 6\}$ and P_i as in Figure 8.

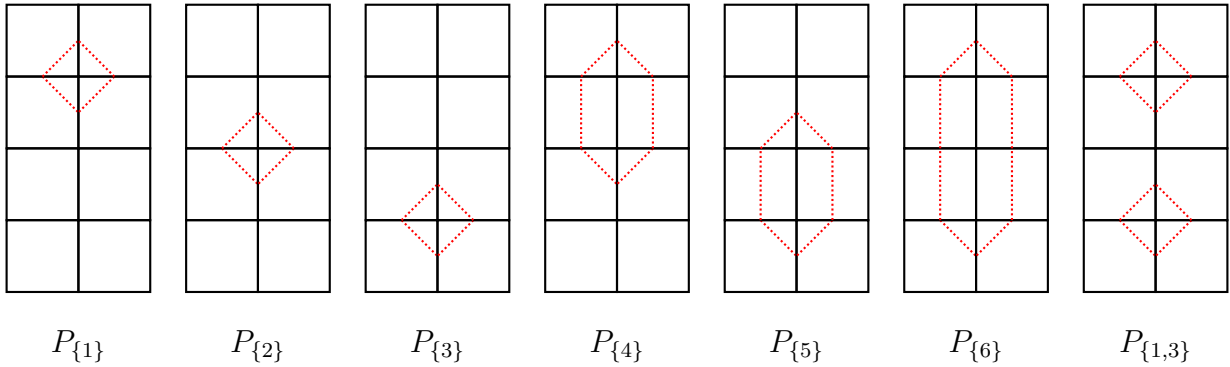


Figure 8: All $(4, 2)$ polygon mosaics except $P_\emptyset$

Next we look at some sets of mosaics. For any $I \subseteq J$ such that P_I exists, let M_I be the set of mosaics that contain the polygons of P_I , i.e., the set of mosaics that can be formed from P_I by using any tile wherever P_I has a blank tile. Notice that the size $|M_I|$ of this set equals 7^a , where a is the number of blank tiles in P_I . By the definition of $\mathcal{L}$, the number of cells of $\mathcal{L}(P_I)$ that are either $\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}$ or $\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}$ is also a , and since $\mathcal{L}(P_I)$ is proper, i.e. it has neither $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ nor $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$ cells, and thus we see that $|v(\mathcal{L}(P_I))| = |M_I|$. Furthermore, $|I|$ is the number of polygons in P_I , so Proposition 4.2 tells us that $v(\mathcal{L}(P_I))$ and $(-1)^{|I|}$ have the

same sign. Therefore, if P_I exists then

$$v(\mathcal{L}(P_I)) = (-1)^{|I|} |M_I|.$$

We must show that the sum $\sum_{\ell} v(\ell)$ over all framed lattices ℓ is equal to the number of polygon free mosaics. The sum is unchanged if we restrict the sum to over all proper lattices ℓ because $v(\ell) = 0$ if ℓ is not proper. Furthermore, since $\mathcal{L}$ gives a bijection from polygon mosaics to proper lattices,

$$\sum_{\ell} v(\ell) = \sum v(\mathcal{L}(P_I)) = \sum (-1)^{|I|} |M_I|,$$

where the second and third sums are over all $I \subseteq J$ such that P_I exists. The total number of mosaics, which happens to be 7^{mn} , equals $|M_{\emptyset}|$. Notice that $J = \emptyset$ if and only if $m = 1$ or $n = 1$. If this is the case then every mosaic is polygon free, the third sum above is $|M_{\emptyset}|$, and our theorem is complete. Therefore we may assume that $m \geq 2$ and $n \geq 2$, so that $J \neq \emptyset$ and the third sum has a nonzero term with $I \neq \emptyset$, for example, $I = \{i\}$ for any $i \in J$.

For any $I \neq \emptyset$ such that P_I exists, $M_I = \bigcap_{i \in I} M_{\{i\}}$ by the definition of the M_I and $M_{\{i\}}$. Thus

$$\sum_{\ell} v(\ell) = |M_{\emptyset}| - \sum (-1)^{|I|-1} \left| \bigcap_{i \in I} M_{\{i\}} \right|,$$

where the sum on the right is over all I such that P_I exists and $\emptyset \neq I \subseteq J$. In this equation, the value on the right side is unchanged if instead we sum over all I such that $\emptyset \neq I \subseteq J$. This is because when P_I is not defined, the P_i , $i \in I$, “overlap” in the sense described above, and so $\bigcap_{i \in I} M_{\{i\}} = \emptyset$. By the inclusion-exclusion principle (see the Appendix), this summation over all I with $\emptyset \neq I \subseteq J$ equals $|\bigcup_{i \in J} M_{\{i\}}|$, i.e., the number of mosaics that contain at least one polygon. Since $|M_{\emptyset}|$ is the total number of mosaics, we conclude, as desired, that $\sum_{\ell} v(\ell)$ is the number of polygon-free mosaics. $\square$

Combining Theorem 4.1 with Theorem 3.2 gives the second main result of our paper.

Theorem 4.3. *Define the 1×1 matrices $A_1 = (7)$, $B_1 = (-1)$, $C_1 = D_1 = (1)$, and for positive integers n , define the $2^n \times 2^n$ matrices*

$$\begin{aligned} A_{n+1} &= \begin{pmatrix} 7A_n & B_n \\ C_n & D_n \end{pmatrix} & B_{n+1} &= \begin{pmatrix} -A_n & B_n \\ 0C_n & D_n \end{pmatrix} \\ C_{n+1} &= \begin{pmatrix} A_n & 0B_n \\ C_n & D_n \end{pmatrix} & D_{n+1} &= \begin{pmatrix} A_n & -B_n \\ C_n & 7D_n \end{pmatrix}. \end{aligned}$$

Then the number of (m, n) polygon free mosaics is the upper left entry of A_n^n .

5 Related Work

Our work can be seen as an extension to Hong and Oh [4] and further generalizing the techniques in Oh et al. [11] to explore a new direction in counting mosaics. Hong and Oh [4] provided bounds on the number of polygon mosaics.

Theorem 5.1 ([4]). *The number of (m, n) polygon mosaics for $m, n \geq 3$ is bounded between $2^{m+n-3} \left(\frac{17}{10}\right)^{(m-2)(n-2)}$ and $2^{m+n-3} \left(\frac{31}{16}\right)^{(m-2)(n-2)}$.*

Our Theorem 3.4 compliments this result with a procedure to exactly count polygon mosaics, as well as exactly recreate the sequence A181245 on the OEIS [6]. Note that unlike here, the authors in [4] did not consider the mosaic made from only blank tiles a polygon mosaic, and so our result differs to theirs by 1. We also point out that our work allows for both polygon and polygon free mosaics to be counted. The methods in [11] could be used to count just polygon mosaics, a fact which is likely known to the authors of [4] and [11], though to our knowledge has not appeared in print.

In related work, Lomonaco and Kauffman [5] introduced mosaics constructed from a tile set of 11 distinct tiles, of which our tile set is a subset. This tile set is diagrammed in Figure 9.

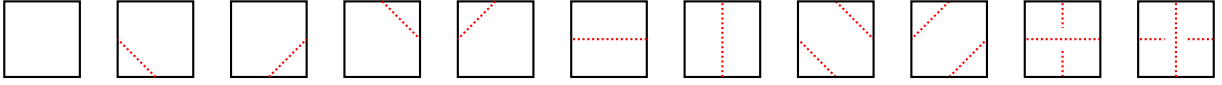


Figure 9: The Lomonaco and Kauffman tile set [5]

The authors were interested in a subset of mosaics which they call *knot mosaics*, as the line segments form knots. Like polygon mosaics, knot mosaics are mosaics in which any tile that is not part of a knot is the blank tile. Oh et al. [11] counted the number of knot mosaics using an analagous method to that of Theorem 3.2. Oh and colleagues go beyond counting by bounding the growth rate of knot mosaics [8, 10, 1], and Oh further adapts techniques similar to Theorem 3.2 to solve problems in monomer and dimer tilings [7, 9]. Related ideas were independently used to count the number of rectangular partitions of a rectangle in [12] for the sequence A182275 on the OEIS [6].

6 Extensions and Further Questions

Counting the number of polygon mosaics and polygon free mosaics in certain extensions of the tile set is simple to do. Throughout this section, fix positive integers m and n . By convention, $\sum_{\ell} v(\ell)$ always designates the sum of $v(\ell)$ over all (m, n) framed lattices ℓ . In all cases, v is a cell product function, and therefore Theorem 3.2 tells us that matrices can be used to evaluate such sums.

Fewer tiles. If we limit ourselves to a subset of the tiles in Figure 1 then, of course, there are fewer polygon mosaics and polygon free mosaics. We can count how many using our existing work. For example, omitting the last tile from left of Figure 1, the number of polygon mosaics equals $\sum_{\ell} v(\ell)$, where v is the cell product function such that $v\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right) = v\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right) = v\left(\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}\right) = v\left(\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}\right) = 0$ and otherwise 1. This can be shown using the reasoning of Theorem 3.1, since the omitted tile is the one that maps via $\mathcal{L}$ to either $\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}$ or $\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}$.

We can similarly count the number of polygon mosaics that can be formed by omitting a different tile or by omitting several tiles. It is worth noting, however, that some of these turn

out to be trivial. Specifically, if the second, third, fourth, or fifth tile from left is omitted from Figure 1 then there is only one possible polygon mosaic, namely the one that contains only blank tiles.

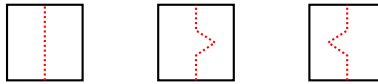
To see this, we outline omitting the second tile; omitting the third, fourth, or fifth tile from left is similar. First, consider the tiles that are not blank, assuming there are some, in a polygon mosaic. Of those tiles, look at those in the highest possible row. The one farthest to the right must be the second tile of Figure 1. Second, in any framed binary lattice in which not all vertices are zero, among the vertices that are 1 and are as high as possible, the one farthest to the right must be the corner of a cell that is $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$. Finally, in Theorem 3.2, if $v(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}) = 0$ and $v(\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}) = 1$ then for all positive integers m and n , the upper right entry of A_n^m equals 1.

As for determining the number of polygon free (m, n) mosaics using a reduced tile set, this will simply be 7^{mn} if the second, third, fourth, or fifth tile is omitted from the tile set, since we have just seen that no polygons can be formed. For these or other omitted tiles, we can use the ideas from Theorem 3.1.

For example, if we omit the last tile of Figure 1 then we use the cell product function v that maps $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$, $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$, $\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}$, and $\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}$ to 0, $\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}$ to 6 (since there are six tiles that can replace the blank tile in a polygon free mosaic), $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}$ to -1 , and the remaining eight cells to 1.

For omitting the blank tile, we would use v defined by mapping $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$ to 0, $\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}$ to 6, $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}$ to -1 , and the remaining ten tiles to 1. In this case we would consider polygon mosaics that might include blank tiles, and create only those polygon free mosaics in which the blank tile was replaced by a tile that is not blank. This example was posed by the first author of this paper and served as the motivation for all our work. The number of polygon free mosaics with this extension is A364440 on the OEIS [6].

Adding tile variations. Adding tiles with the same connection points is similar. For instance, we include three types of tiles that connect the top and bottom midpoints like below:



To count polygon mosaics in this new tile set, take v such that $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$ map to 0, $\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}$ map to 3, and all other cells map to 1. To count polygon free mosaics, take v that maps $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$ to 0, $\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}$ to 3, $\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}$ to 9, $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}$ to -1 , and map 1 to the eight remaining cells.

The Lomonaco and Kauffman tile set. Although we could consider counting the number of knot free mosaics built from the Lomonaco and Kauffman tile set in Figure 9, we will not do so. We invite others to find a way to count such mosaics, but we indicate some potential difficulties. First, there are two reasonable ways to define a knot free mosaic. A knot free mosaic could either be a mosaic in which no knot can be found, or one in which no set of tiles exists for which all segments in those tiles are part of knots. For example, the mosaic

in Figure 10 is polygon free in the second sense but not in the first.

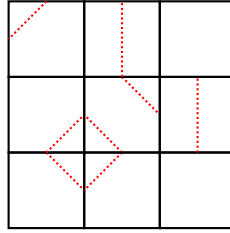


Figure 10: A mosaic that is knot free in the second sense but not the first

A second problem is that the end of the proof of Proposition 4.2 relies on polygons not being able to cross one another, something that is possible in the Lomonaco and Kauffman tile set.

7 Acknowledgments

The authors would like to thank Richard Schank for contributing ideas toward the definition of $\mathcal{L}$ and ideas that led to Theorem 3.2.

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8 Appendix

Proposition (The inclusion-exclusion principle). *For finite sets $A_1, A_2, \dots, A_M$,*

$$\left| \bigcup_{j=1}^M A_j \right| = \sum_{\emptyset \neq I \subseteq \{1, 2, \dots, M\}} (-1)^{|I|-1} \left| \bigcap_{j \in I} A_j \right|.$$

Proof. The $M = 1$ case is trivial, and $M = 2$ case is the familiar and easily justified statement that $|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$. We proceed inductively, so for $M \geq 3$, assume the result for $M - 1$ sets, and let us prove it for M sets. We split up the sum on the right of our desired equation into sets I that do not contain M and those that do. By our inductive hypothesis, the first sum equals $\left| \bigcup_{j=1}^{M-1} A_j \right|$. The second sum is $\sum_{\{M\} \subseteq I \subseteq \{1, 2, \dots, M\}} (-1)^{|I|-1} \left| \bigcap_{j \in I} A_j \right|$, which we rewrite by separating the $I = \{M\}$ term and reindexing the rest using $J = I \setminus \{M\}$ to get

$$|A_M| - \sum_{\emptyset \neq J \subseteq \{1, 2, \dots, M-1\}} (-1)^{|J|-1} \left| \bigcap_{j \in J} (A_j \cap A_M) \right|.$$

Again using our inductive hypothesis, this final sum equals $\left| \bigcup_{j=1}^{M-1} (A_j \cap A_M) \right| = \left| \left(\bigcup_{j=1}^{M-1} A_j \right) \cap A_M \right|$. The entire right side of our desired equation therefore equals $\left| \bigcup_{j=1}^{M-1} A_j \right| + |A_M| - \left| \left(\bigcup_{j=1}^{M-1} A_j \right) \cap A_M \right|$. By the $M = 2$ case, this is $\left| \bigcup_{j=1}^M A_j \right|$, as desired. $\square$